# Writing Ionic Formulae

An ionic solid is formed when positive ion and negative ion react to form a compound. The positive ion and negative ion are attracted to each other because they have opposite charges.

## Rules for Writing Ionic Formulae

1. **Only positive and negative ions will combine.**
   A positive sodium ion, $Na^+$ will combine with a negative chloride ion, $Cl^-$.
   A positive sodium ion, $Na^+$ will **not** combine with a positive calcium ion, $Ca^{2+}$.

2. **The overall charge of the formula is neutral.**
   The formula for sodium chloride is NaCl. The positive charge on the sodium ion is balanced by the negative charge on the chloride ion so the formula is neutral.

3. **The number of each ion is written on the lower right hand side of the ion.**
   The formula for calcium chloride is $CaCl_2$. This shows that for each calcium ion, there are two chloride ions.

4. **Where only one ion is needed, it is not necessary to write the number.**
   When one sodium ion combines with one chloride ion, the formula can be written as NaCl, not $Na_1Cl_1$.

5. **If more than one polyatomic ion is needed, the ion needs to be written in brackets.**
   When a hydroxide ion, $OH^-$, combines with an aluminium ion, $Al^{3+}$, the hydroxide ion needs to be written in brackets. The formula for aluminium hydroxide is $Al(OH)_3$.

**Example 1: Sodium chloride**
Write the charge on the sodium and chloride ions.

Na Cl

For each sodium ion, how many chloride ions do you need to make the charges balanced? Write the formula for sodium chloride below.

**Example 2: Calcium chloride**
Write the charge on the calcium and chloride ions.

Ca Cl

For each calcium ion, how many chloride ions do you need to make the charges balanced? Write the formula for calcium chloride below.

**Example 3: Sodium oxide**

Write the charge on the sodium and oxide ions.

Na O

For each oxide ion, how many sodium ions do you need to make the charges balanced? Write the formula for sodium oxide below.

**Example 4: Magnesium oxide**

Write the charge on the magnesium and oxide ions.

Mg O

For each magnesium ion, how many oxide ions do you need to make the charges balanced? Write the formula for magnesium oxide below.

**Example 5: Aluminium oxide**

Write the charge on the aluminium and oxide ions.

Al O

For each aluminium ion, how many oxide ions do you need to balance the charges? In this situation, you need to find the lowest common multiple of the charges, in this case 6. How many aluminium ions make a charge of +6? How many oxide ions make a charge of -6? Write the formula of aluminium oxide below.

**Example 6: Sodium nitrate**

Write the charge on the sodium and nitrate ions.

Na $NO_3$

For each sodium ion, how may nitrate ions do you need to make the charges balanced? Write the formula for sodium nitrate below. (Do you need to use brackets?)

**Example 7: Magnesium nitrate**

Write the charge on the magnesium and nitrate ions.

Mg $NO_3$

For each magnesium ion, how many nitrate ions do you need to balance the charges?  Write the formula of magnesium nitrate below. (Do you need to use brackets?)